

Atomic Red Test

T1071.001 - Application Layer Protocol: Web

Attackers often leverage web-based protocols (like HTTP/S) for command and control (C2) communication. This is stealthy because it blends with normal network activity, and HTTP requests to benign-looking domains (e.g., `google.com`) are rarely flagged by default.

Hello!

Today, we'll be going over how to install Atomic Red within your test environment!

This test simulates a web-based C2 communication by using PowerShell to send an HTTP GET request:

```
Invoke-WebRequest -Uri "http://www.google.com"
```

This mimics how malware or post-exploitation frameworks might:

- Check for internet connectivity
- Beacon to a remote C2 server
- Download commands or payloads

You'll first need to create a Windows 10 VM from an ISO file and then disable Windows Defender real-time protection.



Virus & threat protection settings

View and update Virus & threat protection settings for Microsoft Defender Antivirus.

Have
Get

Real-time protection

Locates and stops malware from installing or running on your device. You can turn off this setting for a short time before it turns back on automatically.



Real-time protection is off, leaving your device vulnerable.



Off

Help
Give

Change

View
for

Privacy

Privacy

Privacy

Cloud-delivered protection

Provides increased and faster protection with access to the latest protection data in the cloud. Works best with Automatic sample submission turned on.



Cloud-delivered protection is off. [Dismiss](#)
Your device may be vulnerable.



Off

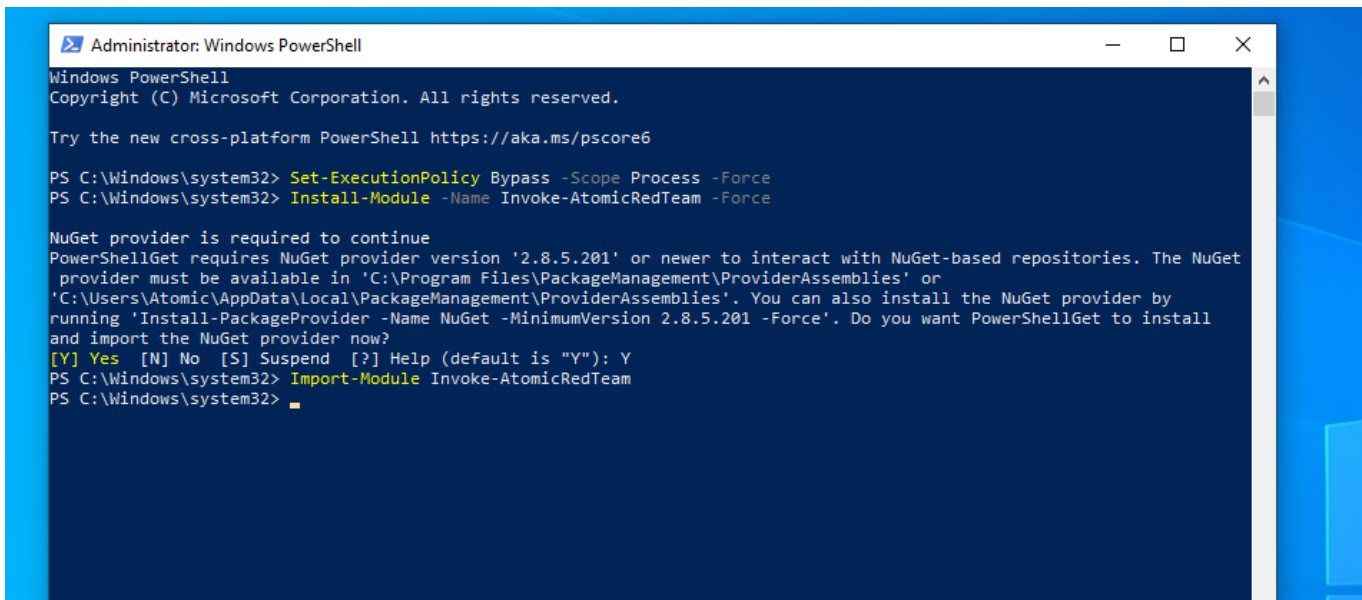
Automatic sample submission

After this has been completed run the following commands so that we may use the Atomic module for Powershell. (<https://www.atomicredteam.io/invoke-atomicredteam>)

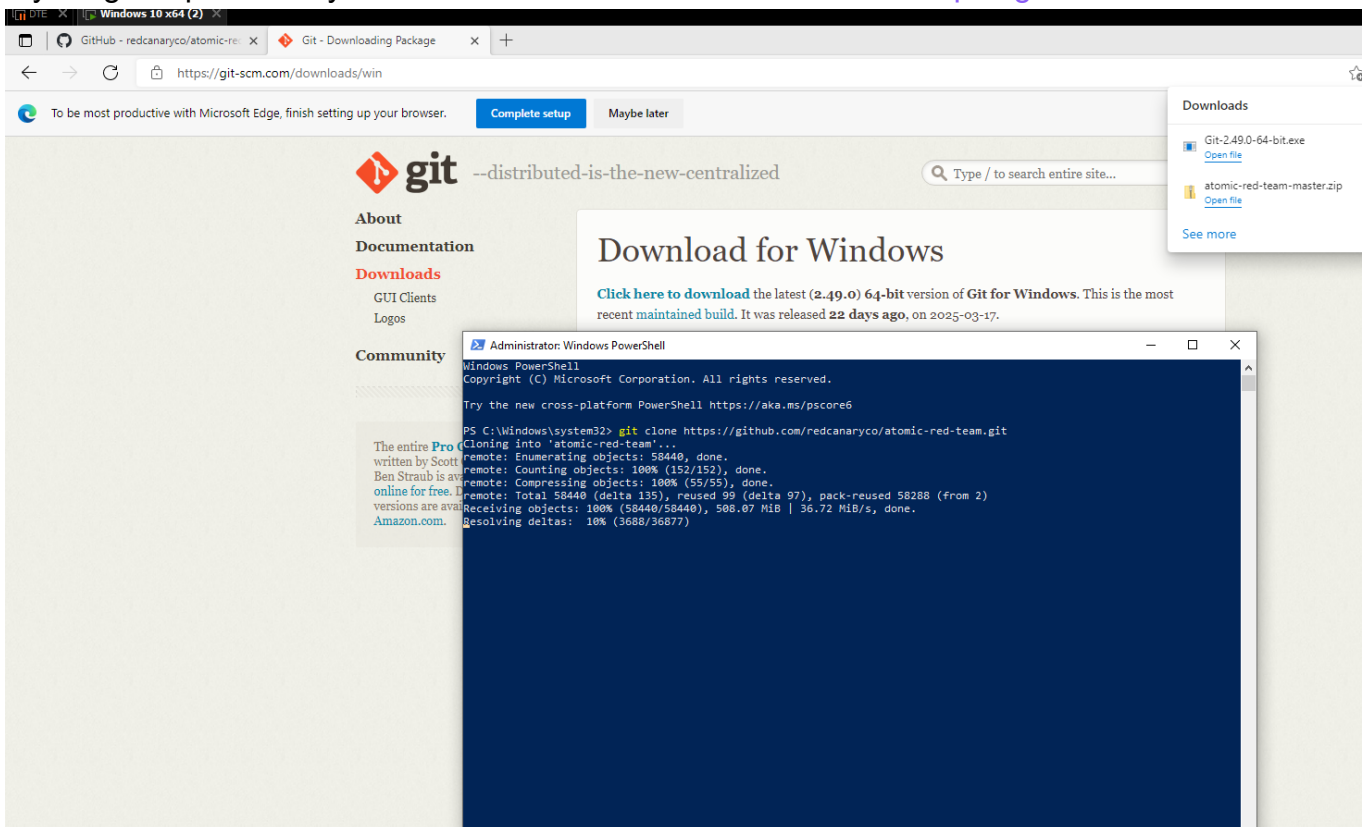
```
Set-ExecutionPolicy Bypass -Scope Process -Force
Install-Module -Name Invoke-AtomicRedTeam -Force
Import-Module Invoke-AtomicRedTeam
```

```
git clone https://github.com/redcanaryco/atomic-red-team.git
```

Set-Location atomic-red-team



If your git request fails you can download the latest version here: <https://git-scm.com/downloads>



Now before we run Atomic we need to download Sysmon using this config:

<https://github.com/SwiftOnSecurity/sysmon-config>

```
sysmon64.exe -accepteula -i sysmonconfig.xml
```

![[Pasted image 20250408120553.png]]

```
Administrator: Windows PowerShell

Mode                LastWriteTime         Length Name
----                -
-a----            4/8/2025  12:09 PM             7490 Eula.txt
-a----            4/8/2025  12:09 PM          8480560 Sysmon.exe
-a----            4/8/2025  12:09 PM          4563248 Sysmon64.exe
-a----            4/8/2025  12:09 PM          4993440 Sysmon64a.exe

PS C:\Users\Atomic\Downloads\Sysmon> .\Sysmon64.exe

System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Usage:
Install:           Sysmon64.exe -i [<configfile>]
Update configuration: Sysmon64.exe -c [<configfile>]
Install event manifest: Sysmon64.exe -m
Print schema:      Sysmon64.exe -s
Uninstall:         Sysmon64.exe -u [force]
  -c Update configuration of an installed Sysmon driver or dump the
      current configuration if no other argument is provided. Optionally
      take a configuration file.
  -i Install service and driver. Optionally take a configuration file.
  -m Install the event manifest (done on service install as well)).
  -s Print configuration schema definition of the specified version.
      Specify 'all' to dump all schema versions (default is latest)).
  -u Uninstall service and driver. Adding force causes uninstall to proceed
      even when some components are not installed.

The service logs events immediately and the driver installs as a boot-start driver to capture activity from early in
the boot that the service will write to the event log when it starts.

On Vista and higher, events are stored in "Applications and Services Logs/Microsoft/Windows/Sysmon/Operational". On
older systems, events are written to the System event log.

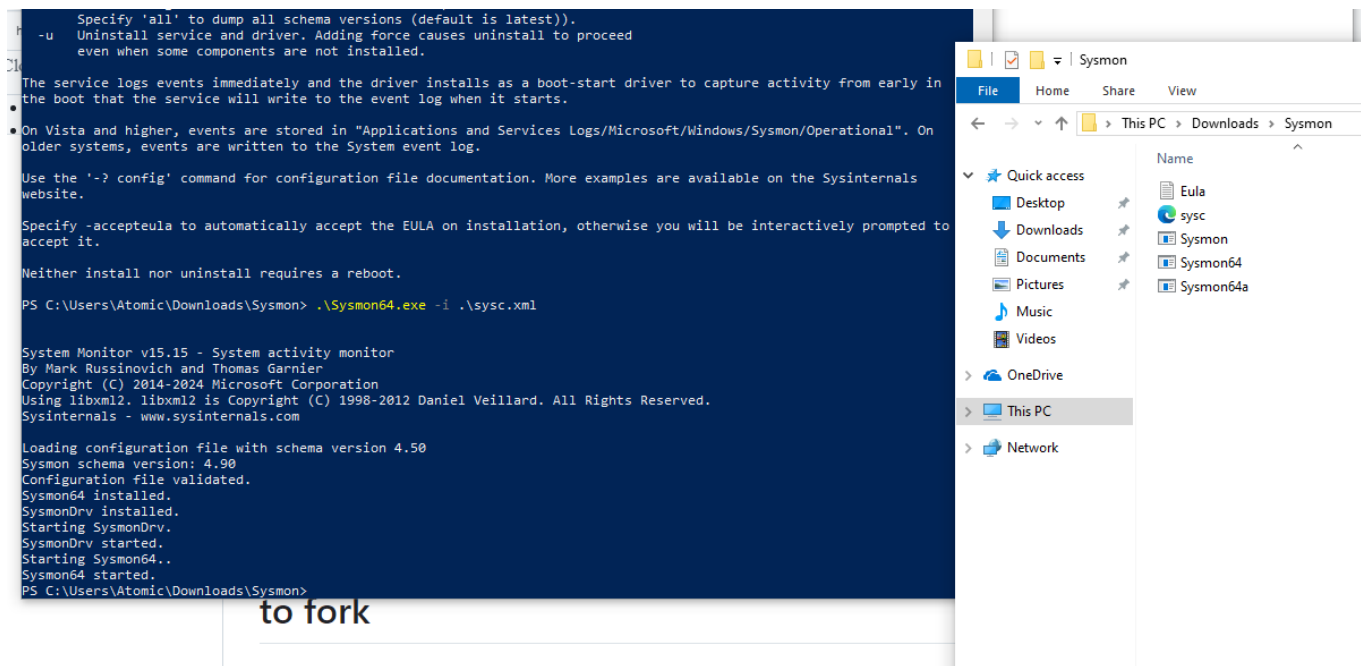
Use the '-? config' command for configuration file documentation. More examples are available on the Sysinternals
website.

Specify -accepteula to automatically accept the EULA on installation, otherwise you will be interactively prompted to
accept it.

Neither install nor uninstall requires a reboot.

PS C:\Users\Atomic\Downloads\Sysmon>
```

>  TerminalService | Key Management Service 68 KB/20 ... 4/8/2025 11:42:50 AM Enabled Overwrite events as nec...



to fork

After installing Sysmon for logging we're then able to execute one of the tactics within Atomic!

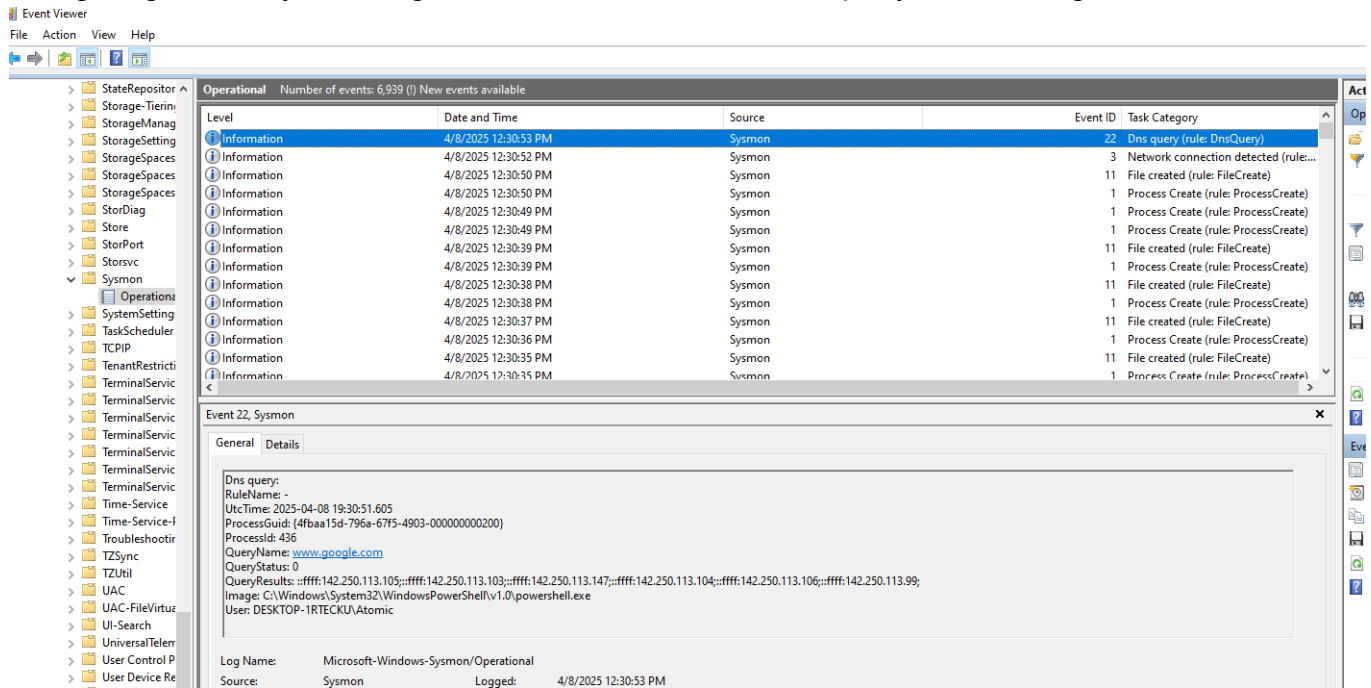
Invoke-AtomicTest T1071.001 -TestNumber 1

```
Administrator: Windows PowerShell
Prereq already met: NanoDump executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\nanodump.x64.exe)
PS C:\Windows\system32> Invoke-AtomicTest T1003.001 -GetPrereqs -Confirm:$false
PathToAtomicFolder = C:\AtomicRedTeam\atomic

GetPrereq's for: T1003.001-1 Dump LSASS.exe Memory using Procdump
Attempting to satisfy prereq: Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\procdump.exe)
Prereq already met: Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\procdump.exe)
GetPrereq's for: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
No Preqs Defined
GetPrereq's for: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
Attempting to satisfy prereq: Dumpert executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\Outflank-Dumpert.exe)
Prereq already met: Dumpert executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\Outflank-Dumpert.exe)
GetPrereq's for: T1003.001-4 Dump LSASS.exe Memory using NanoDump
Attempting to satisfy prereq: NanoDump executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\nanodump.x64.exe)
Prereq already met: NanoDump executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\nanodump.x64.exe)
GetPrereq's for: T1003.001-6 Offline Credential Theft With Mimikatz
Attempting to satisfy prereq: Mimikatz must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\x64\mimikatz.exe)
Prereq already met: Mimikatz must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\x64\mimikatz.exe)
Attempting to satisfy prereq: Lsass dump must exist at specified location (%tmp%\lsass.DMP)
Prereq already met: Lsass dump must exist at specified location (%tmp%\lsass.DMP)
GetPrereq's for: T1003.001-7 LSASS read with pypykatz
Attempting to satisfy prereq: Computer must have python 3 installed
Prereq already met: Computer must have python 3 installed
Attempting to satisfy prereq: Computer must have venv configured at C:\AtomicRedTeam\atomic\..\ExternalPayloads\venv_t1003_001
Prereq already met: Computer must have venv configured at C:\AtomicRedTeam\atomic\..\ExternalPayloads\venv_t1003_001
Attempting to satisfy prereq: pypykatz must be installed
Prereq already met: pypykatz must be installed
GetPrereq's for: T1003.001-8 Dump LSASS.exe Memory using Out-Minidump.ps1
No Preqs Defined
GetPrereq's for: T1003.001-9 Create Mini Dump of LSASS.exe using Procdump
Attempting to satisfy prereq: Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\procdump.exe)
Prereq already met: Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\procdump.exe)
GetPrereq's for: T1003.001-10 Powershell Mimikatz
No Preqs Defined
GetPrereq's for: T1003.001-11 Dump LSASS with createdump.exe from .Net v5
Attempting to satisfy prereq: .Net v5 must be installed
Prereq already met: .Net v5 must be installed
GetPrereq's for: T1003.001-12 Dump LSASS.exe using imported Microsoft DLLs
Attempting to satisfy prereq: Computer must have xordump.exe
Prereq already met: Computer must have xordump.exe
GetPrereq's for: T1003.001-13 Dump LSASS.exe using lolbin rdleakdiag.exe
No Preqs Defined
GetPrereq's for: T1003.001-14 Dump LSASS.exe Memory through Silent Process Exit
Attempting to satisfy prereq: NanoDump executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\nanodump.x64.exe)
Prereq already met: NanoDump executable must exist on disk at specified location (C:\AtomicRedTeam\atomic\..\ExternalPayloads\nanodump.x64.exe)
PS C:\Windows\system32> Invoke-AtomicTest T1071.001 -TestNumber 1
PathToAtomicFolder = C:\AtomicRedTeam\atomic

Executing test: T1071.001-1 Malicious User Agents - Powershell
Exit code: 0
Done executing test: T1071.001-1 Malicious User Agents - Powershell
PS C:\Windows\system32>
```

Navigating to our Sysmon logs we're able to see the DNS query to the Google domain!



Event Viewer

File Action View Help

Operational Number of events: 6,939 (1) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	4/8/2025 12:30:53 PM	Sysmon	22	Dns query (rule: DnsQuery)
Information	4/8/2025 12:30:52 PM	Sysmon	3	Network connection detected (rule: NetworkConnection)
Information	4/8/2025 12:30:50 PM	Sysmon	11	File created (rule: FileCreate)
Information	4/8/2025 12:30:50 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	4/8/2025 12:30:49 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	4/8/2025 12:30:49 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	4/8/2025 12:30:39 PM	Sysmon	11	File created (rule: FileCreate)
Information	4/8/2025 12:30:39 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	4/8/2025 12:30:38 PM	Sysmon	11	File created (rule: FileCreate)
Information	4/8/2025 12:30:38 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	4/8/2025 12:30:37 PM	Sysmon	11	File created (rule: FileCreate)
Information	4/8/2025 12:30:36 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	4/8/2025 12:30:35 PM	Sysmon	11	File created (rule: FileCreate)
Information	4/8/2025 12:30:35 PM	Sysmon	1	Process Create (rule: ProcessCreate)

Event 22, Sysmon

General Details

Dns query:
RuleName:
UtcTime: 2025-04-08 19:30:51.605
ProcessGuid: {4fbaa15d-796a-67f5-4903-000000000200}
ProcessId: 436
QueryName: www.google.com
QueryStatus: 0
QueryResults: ::ffff:142.250.113.105;::ffff:142.250.113.103;::ffff:142.250.113.147;::ffff:142.250.113.104;::ffff:142.250.113.106;::ffff:142.250.113.99;
Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
User: DESKTOP-1RTECKU\Atomic

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Logged: 4/8/2025 12:30:53 PM

More information on the test we just ran: <https://github.com/redcanaryco/atomic-red-team/blob/master/atomics/T1071.001/T1071.001.md>

T1071.001 - Application Layer Protocol: Web Protocols

Description from ATT&CK

Adversaries may communicate using application layer protocols associated with web traffic to avoid detection/network filtering by blending in with existing traffic. Commands to the remote system, and often the results of those commands, will be embedded within the protocol traffic between the client and server.

Protocols such as HTTP/S(Citation: CrowdStrike Putter Panda) and WebSocket(Citation: Brazking-Websockets) that carry web traffic may be very common in environments. HTTP/S packets have many fields and headers in which data can be concealed. An adversary may abuse these protocols to communicate with systems under their control within a victim network while also mimicking normal, expected traffic.

Atomic Tests

- [Atomic Test #1 - Malicious User Agents - Powershell](#)
- [Atomic Test #2 - Malicious User Agents - CMD](#)
- [Atomic Test #3 - Malicious User Agents - Nix](#)

Atomic Test #1 - Malicious User Agents - Powershell

This test simulates an infected host beaconing to command and control. Upon execution, no output will be displayed. Use an application such as Wireshark to record the session and observe user agent strings and responses.

Inspired by APTSimulator - <https://github.com/NextronSystems/APTSimulator/blob/master/test-sets/command-and-control/malicious-user-agents.bat>

Supported Platforms: Windows

auto_generated_guid: 81c13829-f6c9-45b8-85a6-053366d55297

Inputs:

Name	Description	Type	Default Value
domain	Default domain to simulate against	string	www.google.com

Attack Commands: Run with powershell !

```
Invoke-WebRequest #{domain} -UserAgent "HttpBrowser/1.0" | out-null
Invoke-WebRequest #{domain} -UserAgent "Wget/1.9+cvs-stable (Red Hat modified)" | out-null
Invoke-WebRequest #{domain} -UserAgent "Opera/8.81 (Windows NT 6.0; U; en)" | out-null
Invoke-WebRequest #{domain} -UserAgent "**<|>*" | out-null
```



Thank you for coming to my Atomic Talk!